U P T I M E

AI-Assisted On-Demand Technician Dispatch Platform for Industrial, Automobile and Electronics Repair

"Downtime ends here."

Project Report

Bachelor of Engineering — Computer Science Engineering

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Academic Year 2026 – 2027

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1. Executive Summary

UPTIME is an AI-powered, on-demand technician dispatch platform that connects factory owners, vehicle owners, and homeowners with verified repair technicians across three major industries: Industrial & MSME, Automobile, and Electronics & Appliances. The platform uses Artificial Intelligence to diagnose faults before dispatch, match the most suitable technician using a smart scoring algorithm, and track repairs in real time — eliminating unplanned downtime and reducing repair time from days to hours.

India has over 6.3 crore MSME manufacturing units, 30 crore registered vehicles, and nearly every household owns multiple electronic appliances — yet there is no unified digital platform for on-demand industrial or technical repair. UPTIME fills this critical gap with a world-class, AI-driven solution that is India-first, multilingual, and accessible to rural and urban users alike.

69 Features	**141 Database Tables**	**6 Platforms**	**19 World-First Features**
3 Industries	**4 Member Team**	**3 Month Timeline**	**■1,25,000 Cr Market**

2. Problem Statement

Core Problem

When a machine, vehicle, or appliance breaks down, the owner faces a frustrating and costly experience: unreliable phone contacts, long waiting times, no way to verify technician quality, zero cost transparency, and no digital service record. Every hour of downtime is a direct financial loss.

Industry	Problem	Impact
Industrial/MSME	No on-demand technician booking	Production halted for days
Industrial/MSME	No verified skilled technicians	Wrong repairs, more damage
Automobile	No roadside emergency dispatch	Drivers stranded for hours
Automobile	No pre-purchase vehicle scanner	Fraud in used car market
Electronics	No doorstep repair guarantee	Data theft, poor quality
Electronics	No warranty/AMC tracking	Money lost on expired coverage
All Industries	No AI-powered fault diagnosis	Wrong technician dispatched
All Industries	No real-time tracking	No visibility post-booking

Why This Problem Matters

- India has 6.3 crore MSME units — most without in-house engineers or digital repair systems.
- Every hour of machine downtime costs an MSME an estimated ■5,000 – ■50,000 in lost production.
- The used car market in India is 4× the new car market — fraud affects millions annually.
- Over 300 crore electronic devices are in use in India with no unified repair ecosystem.
- No dominant platform exists for on-demand industrial repair — the market is completely unorganised.

3. Market Research & Competitive Analysis

Market Size

Segment	Market Size (India)	Growth
Industrial/MSME Repair	■15,000 Cr	12% CAGR
Automobile Repair	■85,000 Cr	15% CAGR
Electronics Repair	■25,000 Cr	18% CAGR
Total Addressable	■1,25,000 Cr	15% CAGR

Competitive Landscape

Competitor	What They Do	Gap vs UPTIME
Zervician (India)	On-demand industrial booking	No AI, no matching, no tracking
Infinite Uptime	IoT predictive maintenance	Hardware-heavy, not reactive
GoMechanic	Car service booking	No AI diagnosis, no roadside SOS
Urban Company	Home appliance/beauty services	No industrial, no automobile
The Up! App (USA)	Industrial repair marketplace	Not available in India
Cashify	Refurbished device buying/selling	No repair, no technician dispatch

Conclusion: UPTIME operates in a validated but early market — real enough that small competitors exist, but with no dominant incumbent. The platform differentiates through AI-powered diagnosis, smart matching, real-time tracking, and world-first features across three industries simultaneously.

4. Proposed Solution

UPTIME is a two-sided marketplace with an AI layer: asset owners post repair requests, verified technicians accept and complete jobs, and the AI layer makes every interaction smarter — from fault diagnosis to technician matching to post-repair predictions.

Core Value Proposition

- AI diagnoses the fault BEFORE a technician is dispatched — right specialist every time.
- Smart matching algorithm scores technicians on skill, distance, rating, and availability.
- Live GPS tracking of technician en route — full visibility like Uber/Swiggy.
- OTP-based arrival verification — tamper-proof, fraud-proof job lifecycle.
- Tamper-proof digital service certificates with SHA-256 hash verification.
- Multilingual support in 9 Indian languages — voice input for illiterate users.
- Works across Industrial, Automobile, and Electronics — one platform for all repairs.

Core Job Lifecycle

Step	Action	Technology Used
1	Owner reports fault	Text + Voice + Photo + Video + Sound
2	AI fault triage	Gemini 1.5 Flash API
3	Smart technician matching	Weighted scoring algorithm (Java)
4	Owner confirms technician	Ranked list with match score
5	Live tracking begins	Google Maps + WebSocket + GPS
6	OTP arrival verification	OTP + GPS location validation
7	AI Repair Copilot guides tech	Gemini + step-by-step checklist
8	Parts logged, job completed	Digital checklist + signature
9	Rating + tamper-proof certificate	SHA-256 hash + PDF generation
10	Digital twin + prediction update	AI analysis + MySQL update

5. Brand Identity

Element	Detail
Brand Name	UPTIME
Full Title	UPTIME: AI-Assisted On-Demand Technician Dispatch Platform
Tagline	"Downtime ends here."
Secondary	"Fix faster. Run longer."
Hindi/Kannada	"Har machine, har waqt" (Every machine, every time)
Primary Color	Deep Blue #1A3C6E — Trust + Professionalism
Accent Color	Electric Teal #00D4AA — Technology + Energy
Font	Inter — Clean, modern, readable across all devices
Logo Concept	Up arrow embedded in letter U — machine going UP, uptime restored
App Name (Owner)	UPTIME — Book Repairs (Play Store / App Store)
App Name (Tech)	UPTIME Tech — For Verified Technicians
Domain	uptime.in / uptimeapp.in

6. System Architecture

High-Level Architecture

UPTIME follows a multi-tier client-server architecture with a single Spring Boot backend serving all six client platforms through a secured API Gateway. The system is containerised using Docker and deployed on Railway with GitHub Actions CI/CD.

Layer	Components	Purpose
Client Layer	6 platforms (React PWA, Flutter x2, React x3)	User interfaces
API Gateway	Rate limiting + Auth + SSL + Versioning	Single secured entry point
Backend Services	Spring Boot — 8 microservices	Business logic
Data Layer	MySQL + Redis + RabbitMQ	Storage + cache + queue
AI Layer	Gemini 1.5 Flash API	Fault triage + copilot + analysis
External Services	Maps + Razorpay + FCM + Twilio + S3	Maps, payments, notifications
DevOps	Docker + GitHub Actions + Railway	Build, test, deploy
Security	JWT + SSL + AES-256 + Rate limiting	Auth + encryption
CDN	Cloudflare	Global fast delivery

6 Client Platforms

Platform	Type	Stack	Users
Owner Web App	React PWA	React + Tailwind	Factory/car/home owners (desktop)
Owner Mobile App	Flutter App	Flutter + Riverpod	Owners on the go (Android/iOS)
Technician Web App	React Web	React + Tailwind	Technicians — desk use, earnings
Technician Mobile App	Flutter App	Flutter + Riverpod	Technicians — field use (critical)
Admin Web App	React Web	React + Tailwind	Platform team — command center
Backend API	REST + WS	Spring Boot Java	Serves all 5 client platforms

7. Technology Stack

Category	Technology	Reason for Choice
Backend	Java + Spring Boot 3.2	Industry standard, strong ecosystem, VTU curriculum
Owner Web	React 18 + TypeScript	Component-based, PWA support, large ecosystem
Technician Web	React 18 + TypeScript	Consistent stack across web platforms
Admin Web	React 18 + TypeScript	Reusable components, fast dashboard development
Mobile Apps	Flutter 3.x (Dart)	Single codebase, native performance, 60fps
Database	MySQL 8.0	Relational data, spatial indexing, VTU taught
Cache	Redis 7	Session management, rate limiting, fast lookups
Message Queue	RabbitMQ	Async AI calls, notifications, email jobs
AI Engine	Google Gemini 1.5 Flash	Free tier, vision + audio + text, multilingual
Maps	Google Maps API	Best navigation, live tracking, geospatial
Payments	Razorpay	India-first, UPI support, payout API
Push Notifs	Firebase FCM	Free, reliable, cross-platform
SMS/OTP	Twilio	Global OTP delivery, WhatsApp support
File Storage	AWS S3 + Cloudinary	Scalable media storage + image transformation
Video Calls	Agora / WebRTC	Low latency video, free tier available
CI/CD	GitHub Actions	Free for students, automated testing + deploy
Containers	Docker + Compose	Consistent environments, easy deployment
Hosting	Railway	Free tier, simple Spring Boot + React deploy
Monitoring	Sentry	Real-time error tracking across all platforms
State Mgmt Web	Redux Toolkit	Predictable state, DevTools support
State Mgmt App	Riverpod	Flutter-native, better than Provider
HTTP Client Web	Axios + React Query	Caching, retry, loading states built-in
HTTP Client App	Dio	Interceptors, cancel tokens, file upload
UI Library	shadcn/ui + Tailwind	Clean design, fully customisable components
DB Migrations	Flyway	Version-controlled schema changes
API Docs	Springdoc OpenAPI	Auto-generates Swagger UI from annotations
CDN	Cloudflare	DDoS protection, global edge caching

8. Industrial & MSME Features

The Industrial module targets small and mid-sized manufacturing units (MSMEs), factories, and workshops. These businesses operate heavy machinery such as CNC lathes, conveyors, hydraulic presses, motors, and pumps. A single machine breakdown can halt an entire production line, causing significant financial loss. UPTIME brings a digital, AI-assisted repair ecosystem to this largely unorganised sector.

#	Feature	Why It Is Used
1	Fault Reporting (Multi-input)	Factory workers report faults via text, voice, photo, video, or sound recording — accommodating varying literacy levels and working conditions.
2	AI Fault Triage	Gemini API analyses the input and returns the likely fault category, urgency level, estimated cost, and required skill tags — ensuring the right specialist is dispatched.
3	Smart Technician Matching	Scores every available technician on weighted distance (40%) and skill match (60%), then ranks them — much smarter than a simple nearest-available approach.
4	Live Job Tracking	Google Maps shows the technician moving in real time, with live ETA updates — eliminates the "where is my technician?" problem entirely.
5	OTP Arrival Verification	A one-time password is sent to the owner when the technician arrives; entering it unlocks the job — prevents fake arrivals and fraudulent job claims.
6	On-Site Checklist	Step-by-step digital checklist guides the technician through the repair — ensures no steps are missed and creates a structured service record.
7	Ratings & Trust System	Two-way ratings (owner rates technician AND technician rates factory) build a reputation economy that improves match quality over time.
8	Service History Dashboard	Every repair is permanently stored against the machine — owners can see total spend, fault frequency, and repair patterns at a glance.
9	Cost Estimator	AI provides a transparent cost range before booking — owners know what to expect and can make informed decisions without surprises.
10	Technician Verification	Document upload, AI face matching, and admin approval ensure only qualified, identity-verified technicians can accept jobs.
11	Admin Analytics Dashboard	Real-time platform metrics, live job map, technician performance, and fault heatmaps allow the platform team to manage operations at scale.
12	Repair Time Auction	For urgent breakdowns, owners post a bonus; nearby technicians race to arrive first — dramatically reduces emergency response time.

Industrial Unique / World-First Features

#	Unique Feature	Why World-First
U1	AI Repair Copilot	AI guides technician step-by-step on screen during repair — like GPS navigation but for fixing machines. No competitor does this.
U2	Machine Sound Analysis	Owner records machine sound; AI analyses audio patterns and identifies the fault — no photo or description needed. Works for illiterate workers.
U3	Digital Twin of Machine	AI creates a virtual copy of every machine, tracks component health over time, and predicts which part will fail next — without any IoT sensors.

#	Unique Feature	Why World-First
U4	AI Negotiation Assistant	AI acts as a fair price referee — suggests market-rate pricing, flags overcharging, and guides counter-offer negotiation between owner and technician.
U5	Repair Pattern Prediction	AI analyses historical repair data across the platform and predicts future breakdowns weeks in advance — sensorless predictive maintenance.
U6	Tamper-Proof Job Records	SHA-256 cryptographic hash locks every job record permanently — legal-grade proof of service that cannot be edited or disputed.
U7	OTP + GPS Arrival Verify	Combining OTP (physical presence) with GPS location validation makes arrival verification bulletproof — two-layer fraud prevention.
U8	Insurance Per Job	Every job auto-generates a micro-insurance policy — first repair platform to offer built-in job insurance for Indian MSMEs.
U9	Skill NFT Badges	Technicians earn blockchain-verified skill badges — portable credentials they own forever, even if they leave the platform.
U10	Crowd-Sourced Repair Wiki	Every repair automatically contributes knowledge to a living encyclopedia — AI extracts insights and builds the world's most accurate repair database.
U11	Virtual Repair Consultation	Owner video-calls a technician for live remote diagnosis — AI overlays fault annotations on the live feed. Physical visit only if necessary.

9. Automobile Features

The Automobile module targets car owners, bike owners, truck drivers, and workshop operators across India. With over 30 crore registered vehicles and a largely unorganised roadside repair market, UPTIME brings Uber-style emergency dispatch, AI vehicle diagnosis, and complete digital vehicle management to every driver.

#	Feature	Why It Is Used
1	Roadside SOS Emergency	One-tap emergency dispatch sends the nearest verified mechanic with live tracking — eliminates the terrifying experience of being stranded.
2	Vehicle Fault Reporting	Multi-input fault reporting including dashboard warning light photo analysis — AI reads the warning light and tells the driver exactly what is wrong.
3	Smart Workshop Booking	BookMyShow-style slot booking at verified garages — owners see ratings, services, and available time slots and confirm instantly.
4	Vehicle Digital Twin	Complete digital life of every vehicle — service history, parts replaced, fuel logs, health score, and AI-predicted next maintenance.
5	Roadside Auction	Mechanics compete to reach the stranded driver fastest — gamified emergency response with real-time race tracking on a live map.
6	EV Specific Support	Dedicated EV diagnosis, certified EV mechanics, battery health tracking, charging station finder, and mobile charging van dispatch.
7	Insurance Claim Assistant	AI guides the driver step-by-step at the accident spot — collects evidence, auto-fills claim forms, and tracks settlement in real time.
8	Tyre and Fuel Service	Mobile tyre repair van and emergency fuel delivery dispatched to the driver's location — fixes the two most common roadside problems.
9	Driver Safety Score	Phone sensors analyse braking, acceleration, cornering, and phone usage during every trip — safe drivers earn insurance discounts and platform rewards.
10	Fleet Management	Taxi companies and logistics operators manage entire vehicle fleets — track maintenance schedules, driver scores, and repair costs in one dashboard.
11	Traffic Fine Manager	View, pay, and contest e-challans — vehicle document expiry reminders for insurance, PUC, and driving licence keep drivers legally compliant.
12	Fuel Station Locator	Real-time map of petrol, diesel, CNG, and EV charging stations — integrated with vehicle fuel type for instant relevant results.

Automobile Unique / World-First Features

#	Unique Feature	Why World-First
UA1	Engine Sound Fingerprint	AI learns YOUR specific engine's unique sound baseline — detects 0.1% deviation weeks before any visible symptom appears. No platform does this.
UA2	Vehicle Mood Score	AI analyses 50+ parameters (engine, tyres, battery, weather, driving history) to give a daily "mood score" — tells if the car is safe to drive today.
UA3	Pre-Purchase Vehicle Scanner	Independent AI-powered 200-checkpoint inspection before buying a used car — detects odometer fraud, flood damage, accident history, and pending bank loans.

10. Electronics & Appliances Features

The Electronics module covers both home appliances (AC, refrigerator, washing machine, geyser, microwave, TV) and personal electronics (smartphones, laptops, tablets). Every Indian household owns multiple devices, yet the repair ecosystem remains plagued by fake service centres, data theft, opaque pricing, and long waiting times.

#	Feature	Why It Is Used
1	Smart Device Fault Reporting	Multi-input diagnosis including error code lookup and DIY fix suggestions — often saves the owner a service call entirely.
2	Repair vs Replace Advisor	Unbiased AI analysis tells owners honestly whether repair is financially worth it — prevents unnecessary replacement and unnecessary repair.
3	Doorstep Repair Service	Verified technician visits the home at a convenient time slot — repair done in front of the owner with data protection protocols active.
4	Device Health Monitor	AI tracks health scores for all home devices, correlates electricity bill changes with appliance performance, and sends seasonal maintenance alerts.
5	Data Protection Protocol	Guided pre-repair data backup, guest mode activation, AI-monitored repair session, and post-repair data integrity certificate — eliminates data theft risk.
6	Spare Parts Marketplace	Verified genuine parts marketplace with AI part finder, fast delivery, warranty tracking, and eco-friendly refurbished options.
7	Warranty & AMC Manager	All warranties and Annual Maintenance Contracts in one place — AI reads warranty cards via OCR, tracks expiry, and guides claims automatically.
8	Smart Home Integration	Connects to Google Home, Alexa, Samsung SmartThings — AI monitors IoT data and raises repair requests automatically when anomalies are detected.
9	Electronics Insurance	Comprehensive device insurance covering accidental damage, theft, liquid damage, and breakdown — AI processes claims instantly with cashless repair.
10	Certified Refurbished Market	Technician-certified refurbished devices with grade A/B/C transparency, IMEI theft check, data wipe certificate, and 6-month warranty.
11	B2B Electronics Repair	Office IT support contracts for businesses — bulk repair plans, asset tracking, and dedicated technician assignments for corporate clients.
12	Student Device Care Plan	School and college bulk repair plans at subsidised rates — designed for educational institutions managing large numbers of student devices.

Electronics Unique / World-First Features

#	Unique Feature	Why World-First
UE 1	Fault Epidemic Detection	AI monitors all repair reports in real time; when 50+ users report the same fault on the same model simultaneously, an epidemic alert is triggered with manufacturer notification and class-action support.
UE 2	Silent Killer Detection	Monitors home appliances for invisible dangers: refrigerant gas leaks, electrical leakage, microwave radiation, carbon monoxide, and fire risk — alerts family before harm occurs.

#	Unique Feature	Why World-First
UE 3	Crowd-Sourced Repair Wiki	Every completed repair auto-contributes to a living knowledge base — AI extracts repair steps, success rates, and cost data. The world's first repair encyclopedia built from real job data.
UE 4	Universal Repair Language	Supports all 22 Indian official languages, 1,600+ dialects, voice-only mode for illiterate users, WhatsApp/SMS fallback for feature phones, and Indian Sign Language for deaf users.
UE 5	Device End of Life Ritual	When a device is beyond repair: AI generates a memory reel, guides complete data migration, creates an eco-disposal certificate, and helps find the perfect replacement — an emotional and practical farewell experience.

11. Platform-Wide Unique Features

Beyond industry-specific features, UPTIME includes several cross-industry innovations that apply to all three sectors simultaneously.

Feature	Applies To	Description
Live Video Consultation	All 3	Owner video-calls a technician before booking. AI overlays fault annotations on the live feed. Physical visit only if remote fix is not possible. Dramatically reduces unnecessary dispatches.
Technician NFT Badges	All 3	Every proven skill earns a blockchain-verified badge that the technician owns permanently — portable credentials that survive platform changes.
Repair Carbon Credits	All 3	Every repair instead of replacement earns carbon credits. Credits are tradeable or donatable. First repair platform with a built-in carbon credit economy.
Subscription & Loyalty	All 3	Monthly plans for owners (Basic/Standard/Premium) with job limits, priority matching, and discounts. Referral bonuses for both owners and technicians.
In-App Payment Gateway	All 3	UPI, cards, EMI, and UPTIME wallet — all payment methods supported natively. Razorpay powers transactions and automated technician payouts.
Technician Earnings Dash	All 3	Complete income dashboard with GST records, TDS calculations, monthly summaries, and one-tap bank withdrawal via UPI.
AI Chatbot Support 24/7	All 3	AI chatbot answers platform questions, creates support tickets, and escalates complex issues to human agents — available in all supported languages.

12. Platform Design — 6 Platforms

Platform	Users	Key Screens / Features
Owner Web App (React PWA)	Factory/car/home owners on desktop	Landing, dashboard (industry switcher), fault report (voice+photo+video+sound), AI diagnosis, technician ranked list, live tracking map (Uber-style), job history, digital twin, insurance vault, warranty manager, marketplace, video consult, subscription
Owner Mobile App (Flutter)	Owners on Android/iOS	Same as web app optimised for mobile — bottom navigation, full-screen map, bottom sheets, push notifications, offline mode, voice input in regional languages, SOS emergency button
Technician Web App (React)	Technicians at home/workshop	Profile setup, skill tagging, document upload, earnings dashboard with charts, job history table, availability calendar, badges collection, learning center (Wiki), workshop team management, analytics
Technician Mobile App (Flutter)	Technicians on field (critical)	Home with availability toggle, incoming job request card (accept/decline), Google Maps navigation, OTP entry, AI copilot step guide, photo capture, parts logging, digital signature, job complete animation, earnings preview
Admin Web App (React)	Platform team	Live platform dashboard, technician verification queue, live jobs map, dispute resolution center, epidemic monitor, silent killer alerts, finance dashboard, mass notification composer, AI prompt editor, feature flags, platform settings
Spring Boot Backend	Serves all 5 platforms	8 service modules: Auth, Job, AI, Matching, Payment, Notification, Tracking, Admin — REST APIs + WebSocket — serves 141 DB tables across MySQL, Redis, RabbitMQ

UI Design References

Reference	What We Take From It
Urban Company	Pure white background, generous whitespace, subtle card shadows, clean typography, verified badges, trust signals, category icon grid
Uber	Full-screen map with bottom sheet overlay, real-time marker movement, ETA card, minimal UI on top of map, driver/tech info card
Ajio	Rich product cards, bold typography hierarchy, color accents used sparingly, smooth scroll, strong call-to-action buttons

13. Complete Database Schema

The UPTIME database consists of 141 tables organised into 14 functional modules. The schema uses MySQL 8.0 with spatial indexing for geolocation queries, Flyway for version-controlled migrations, and soft deletes throughout for data integrity and legal compliance.

Module	Tables	Key Tables
1. Users & Auth	8	users, user_addresses, user_sessions, otp_verifications, user_preferences, referrals
2. Technicians	10	technicians, technician_skills, technician_documents, technician_locations, technician_badges, technician_bank_accounts
3. Jobs & Workflow	12	jobs, job_media, job_checklist, job_parts_used, job_disputes, job_auction, job_otp_verifications
4. AI & Diagnosis	12	ai_fault_diagnoses, digital_twins, digital_twin_components, repair_predictions, ai_copilot_sessions, epidemic_detections, silent_killer_alerts
5. Matching	6	matching_requests, matching_scores, technician_dispatch_history, surge_pricing, live_location_tracking
6. Payments	10	transactions, wallet_transactions, payouts, subscriptions, invoices, gst_records, refunds, platform_revenue
7. Ratings	7	job_ratings, technician_trust_scores, owner_trust_scores, platform_nps, rating_analytics
8. Industry Specific	12	machines, vehicles, vehicle_documents, vehicle_fuel_logs, driver_safety_scores, pre_purchase_scans, devices, device_health_logs, smart_home_connections
9. Insurance	5	insurance_policies, insurance_claims, warranties, amc_contracts, extended_warranties
10. Marketplace	6	marketplace_products, marketplace_sellers, marketplace_orders, marketplace_refurbished, spare_parts_requests
11. Notifications	5	notifications, notification_templates, notification_preferences, mass_notifications
12. Analytics & Logs	6	platform_analytics, api_logs, error_logs, audit_logs, user_activity_logs
13. Wiki & Knowledge	5	wiki_articles, wiki_versions, wiki_contributions, wiki_votes, wiki_comments
14. Admin & Platform	7	admin_users, admin_roles, platform_settings, feature_flags, cookie_consents, end_of_life_records
TOTAL	141	1,700+ columns · 500+ indexes · 8 geospatial tables · 15 soft-delete tables

Key Table Structures

Table: jobs (Core table — 35 columns)

Column	Type	Purpose
id	BIGINT PK	Primary identifier
job_number	VARCHAR UNIQUE	Human-readable ID (UPT-2026-00234)
owner_id	FK → users	Who posted the job
technician_id	FK → technicians	Assigned technician (nullable)
industry	ENUM	INDUSTRIAL / AUTOMOBILE / ELECTRONICS
status	ENUM	POSTED → MATCHED → EN_ROUTE → IN_PROGRESS → COMPLETED
owner_location	POINT	Geospatial — enables distance-based matching
ai_fault_category	VARCHAR	Output of Gemini diagnosis
estimated_cost_min/max	DECIMAL	AI cost estimate range
final_cost	DECIMAL	Agreed cost after completion
hash_value	VARCHAR(64)	SHA-256 tamper-proof lock
is_tamper_proof_locked	BOOLEAN	True after job completion — immutable
scheduled_at	DATETIME	For planned/preventive jobs
completed_at	DATETIME	Timestamp for analytics

Table: technicians (Core table — 25 columns)

Column	Type	Purpose
id	BIGINT PK	Primary identifier
user_id	FK → users	Links to base user account
industry_type	ENUM	INDUSTRIAL / AUTOMOBILE / ELECTRONICS / ALL
is_available	BOOLEAN	Online/offline toggle — live status
verification_status	ENUM	PENDING / APPROVED / REJECTED / SUSPENDED
rating_avg	DECIMAL(3,2)	Computed field — updated after every rating
wallet_balance	DECIMAL	Earnings available for withdrawal
preferred_language	ENUM	EN / HI / KN / TA / TE — for multilingual UI

Table: matching_scores (Algorithm output)

Column	Type	Purpose
total_score	DECIMAL	Final weighted score (0–100)
distance_score	DECIMAL	Higher for closer technicians

Column	Type	Purpose
skill_score	DECIMAL	How well skills match AI diagnosis
distance_km	DECIMAL	Actual distance calculated
eta_mins	INT	Estimated arrival time
rank_position	INT	1 = best match shown first to owner
response	ENUM	PENDING / ACCEPTED / DECLINED / EXPIRED

Database Design Principles

- Soft deletes on all user-facing tables (deleted_at column) — data never permanently deleted for legal compliance.
- Geospatial POINT columns with SPATIAL INDEX on location tables — enables efficient "find nearest technician" queries.
- SHA-256 hash values on jobs, invoices, and pre-purchase scan tables — cryptographic tamper-proofing.
- Flyway version-controlled migrations (V1__... to V17__...) — team can sync schema changes safely.
- JSON columns for flexible arrays (skill_tags, fault_indicators, step_data) — avoids over-normalisation.
- Partitioning on high-volume tables (jobs, api_logs, location_tracking) — performance at scale.
- Database views for common dashboard queries — reduces repeated complex JOIN logic.
- Triggers for automatic statistic updates — technician rating_avg auto-recalculates after each job.

14. Project Folder Structure

GitHub Repository Layout

The project is organised as a monorepo with six separate codebases, a shared types package, and a documentation folder. Each codebase has its own Dockerfile and is deployed independently.

Repository / Folder	Type	Description
uptime-backend/	Spring Boot	Java backend — all 8 service modules, 14 DB migration files, AI prompt templates, schedulers
uptime-owner-web/	React PWA	Owner web app — 30+ pages, 50+ components, Redux store, i18next multilingual, PWA service worker
uptime-owner-app/	Flutter	Owner mobile app — 20+ screens, Riverpod state, Google Maps, voice input, offline Hive storage
uptime-tech-web/	React	Technician web app — dashboard, earnings, badges, workshop management, analytics
uptime-technician-app/	Flutter	Technician field app — job requests, navigation, AI copilot, checklist, parts logging, earnings
uptime-admin-web/	React	Admin command center — live map, verification queue, dispute resolution, epidemic monitor
uptime-shared/	TypeScript	Shared types, constants, and utility functions used across all three React apps
uptime-docs/	Markdown /PDF	API documentation, database schema docs, user guides, architecture diagrams, project report
.github/workflows/	YAML	CI/CD pipelines — backend, owner-web, tech-web, admin-web, Flutter all have separate workflows
docker-compose.yml	Docker	Runs all services locally: MySQL, Redis, RabbitMQ, backend, owner-web, tech-web, admin-web

Backend Module Structure

Module	Package	Contains
Auth	modules/auth/	AuthController, OTPService, RefreshTokenService, JWT config
User	modules/user/	UserController, UserService, User entity, Address entity
Technician	modules/technician/	TechnicianController, SkillService, VerificationService, LocationService
Job	modules/job/	JobController, JobService, ChecklistService, DisputeService, AuctionService
AI	modules/ai/	GeminiService, FaultTriageService, CopilotService, EpidemicService, PredictionService
Matching	modules/matching/	MatchingService, ScoringEngine, DispatchService, SurgePricingService

Module	Package	Contains
Payment	modules/payment/	RazorpayService, WalletService, PayoutService, InvoiceService, GSTService
Notification	modules/notification/	FCMService, SMSService (Twilio), EmailService (SendGrid), TemplateService
Tracking	modules/tracking/	WebSocketHandler, LocationService, LiveTrackingService
Industry	modules/industry/	MachineService, VehicleService, DeviceService — industry-specific logic
Insurance	modules/insurance/	InsuranceService, ClaimService, WarrantyService, AMCService
Marketplace	modules/marketplace/	ProductService, OrderService, SellerService, RefurbishedService
Wiki	modules/wiki/	WikiService, ContributionService — knowledge base management
Admin	modules/admin/	AdminService, PlatformAnalyticsService, FeatureFlagService, DisputeResolutionService
Schedulers	scheduler/	RepairPredictionScheduler, EpidemicScheduler, ReminderScheduler, PayoutScheduler

15. Security Architecture

Security Layer	Implementation	Protection Against
Authentication	JWT (RS256) + OTP	Unauthorised access, session hijacking
Authorisation	Spring Security RBAC	Privilege escalation, cross-user data access
Transport Security	HTTPS + TLS 1.3	Man-in-the-middle, eavesdropping
Data Encryption	AES-256 at rest	Data breach exposure of sensitive fields
Input Validation	Bean Validation + Zod	SQL injection, XSS, malformed input
CORS Policy	Whitelist origins only	Cross-site request forgery
Rate Limiting	Redis token bucket	Brute force, DoS attacks
Content Security Policy	CSP headers React apps	XSS via injected scripts
File Upload Validation	MIME + size + magic bytes	Malware upload, oversized files
Brute Force Protection	Redis attempt counter	Password and OTP guessing attacks
API Key Security	Env vars + rotation plan	API key exposure in source code
Audit Logging	Immutable audit_logs table	Insider threats, compliance
OWASP Top 10	Addressed in design	All major web vulnerability categories
Admin Security	2FA + IP whitelist	Admin account compromise
Tamper-Proof Records	SHA-256 job hashes	Post-completion record manipulation

Legal & Compliance

Requirement	How UPTIME Addresses It
India PDPB 2023	Cookie consent banner, data deletion flow, explicit consent for sensitive data, privacy policy page on all platforms
GDPR (for global use)	Right to be forgotten (30-day deletion), data portability, consent records in cookie_consents table
GST Compliance	gst_records table stores CGST/SGST/IGST per transaction; GST reports downloadable by technicians
Terms of Service	Displayed on signup for both owners and technicians; digital acceptance recorded with timestamp
Age Verification	Date of birth required on signup; under-18 users blocked from platform
Technician Contract	Digital agreement signed at onboarding; stored in technician_documents table
Data Breach Response	Defined incident response plan; Sentry monitors for anomalies; admin alerted within 15 minutes

16. DevOps & Deployment

CI/CD Pipeline

Every push to any branch triggers the GitHub Actions pipeline — tests run automatically and successful builds on the main branch are deployed to Railway without manual intervention.

Stage	Trigger	Actions
Lint & Test	Every push	Run JUnit tests, ESLint, Dart analyser
Build	Every push	Maven package, npm build, Flutter build APK
Docker Build	Push to develop	Build Docker images for all services
Deploy Staging	Merge to develop	Auto-deploy to Railway staging environment
Deploy Prod	Merge to main	Auto-deploy to Railway production after approval
Notify	On success/failure	Slack/email notification to team

Infrastructure

Service	Hosting	Tier
Spring Boot API	Railway	Free tier (upgradeable)
MySQL 8.0	Railway	Managed database
Redis 7	Railway	Redis plugin
RabbitMQ	Railway	Plugin
Owner Web App	Vercel/Railway	Free tier
Tech Web App	Vercel/Railway	Free tier
Admin Web App	Vercel/Railway	Free tier
Static Assets	Cloudflare CDN	Free tier
Media Files	AWS S3	Pay per use
Error Monitoring	Sentry	Free tier
Uptime Monitor	UptimeRobot	Free tier

17. Development Roadmap

Phase	Duration	Deliverables	Success Criteria
Phase 1 — Foundation	Week 1–2	DB schema, Spring Boot setup, React project init, design system	All tables created, APIs return 200, UI renders
Phase 2 — Core Flow	Week 3–4	Auth, job posting, AI diagnosis, technician list	Owner can post fault and see AI result
Phase 3 — AI Integration	Week 5	Gemini fully integrated, matching algorithm, real-time tracking	End-to-end job flow working
Phase 4 — Flutter Apps	Week 6–7	Owner app and Technician app built	Technician can accept and complete job on phone
Phase 5 — Payments	Week 8	Razorpay, wallet, invoices, GST	Full payment flow working end-to-end
Phase 6 — Unique Features	Week 9–10	Sound analysis, digital twin, epidemic detection, auctions	5+ unique features live and demonstrable
Phase 7 — Polish	Week 11	Loading states, error pages, multilingual, offline mode	App feels production-quality
Phase 8 — Testing	Week 11	Unit tests, integration tests, load tests, security scan	80%+ test coverage on critical paths
Phase 9 — PBL Prep	Week 12	Demo script, seed data, presentation deck, report final	Full demo rehearsal successful

MVP Definition (End of Month 1)

- Owner can sign up, login, and post a fault with photo/text.
- Gemini API analyses the fault and returns diagnosis with cost estimate.
- Matching algorithm scores and ranks available technicians.
- Technician receives job request, accepts, and navigates to location.
- OTP arrival verification and basic checklist working.
- Job marked complete with rating submitted.
- Basic admin dashboard showing live jobs.

18. Team Roles & Division of Work

The project is developed by a team of four Computer Science Engineering students from DSATM, Bengaluru. Work is divided based on skill level and module boundaries.

Member	Role	Primary Responsibilities
Member 1 (Tech Lead)	Full Stack + AI	Owner web app (React), AI/Gemini integration, matching algorithm, WebSocket server, system architecture, code reviews, deployment
Member 2	Backend + Database	MySQL schema + Flyway migrations, Spring Boot APIs (Auth/Job/Payment/Admin), Razorpay integration, Redis setup, security implementation
Member 3	Frontend + Mobile	Technician web app (React), both Flutter apps (owner + technician), Admin web frontend, multilingual setup, UI/UX polish
Member 4	Testing + Documentation	Manual testing all 69 features, Postman API tests, project report, PBL presentation deck, demo data creation, bug tracking

Git Workflow

- main branch — production only; only Tech Lead merges here.
- develop branch — staging; all feature branches merge here after review.
- feature/member-name/feature-name — individual feature branches.
- Pull Request required for every merge; minimum 1 approval before merge.
- Daily standup at 10 AM — 30 minutes — what done, what next, any blockers.

19. Future Scope

Feature	Timeline	Description
Native Payment App	6 months	UPTIME Pay — escrow-style payment release on job completion with UPI integration
IoT Sensor Kit	1 year	UPTIME Sensor — small hardware device that attaches to any machine and sends real-time health data to the platform
Predictive ML Model	6 months	Train a custom ML model on completed job data for more accurate fault prediction than prompt-based AI
Government Partnership	1 year	Share anonymised fault heatmap data with Smart City Mission and MSME Ministry for policy making
Insurance Company API	6 months	Direct API integration with Acko, Digit Insurance for real policy issuance and instant cashless claims
Carbon Credit Exchange	1 year	Partner with carbon credit platforms to make UPTIME repair credits tradeable for real monetary value
Blockchain NFT Badges	1 year	Deploy smart contracts on Polygon blockchain to mint technician skill badges as true NFTs
UPTIME Academy	1 year	Full training and certification platform for technicians — partnered with ITI and polytechnic colleges
Pan-India Expansion	2 years	Launch in all Tier 1 and Tier 2 cities with local technician recruitment drives
International Expansion	3 years	South-East Asia (Indonesia, Vietnam) — similar MSME-heavy markets with same problem statement
Autonomous Diagnosis Robot	5 years	AI-powered mobile robot that visits factories and diagnoses machines autonomously without a human technician

20. Conclusion

UPTIME represents a comprehensive, technically rigorous, and commercially viable solution to one of India's most persistent yet overlooked problems: the absence of a reliable, digital, AI-powered repair ecosystem.

The platform is differentiated from all existing competitors through its combination of AI-powered fault diagnosis before dispatch, smart skill-based technician matching, Uber-style real-time tracking, tamper-proof service records, and 19 world-first features that no competitor has attempted anywhere in the world.

The architecture is production-grade: 141 database tables, 6 client platforms, a secured API gateway, Redis caching, async message queues, geospatial indexing, SHA-256 tamper proofing, end-to-end encryption, and a complete CI/CD pipeline. This is not merely a student project — it is a fully designed and specifiable startup product.

UPTIME targets a combined addressable market of ■1,25,000 crore across three industries in India alone. With 6.3 crore MSMEs, 30 crore vehicles, and hundreds of crore electronic devices in daily use — every one of them will break down at some point. When they do, UPTIME will be there.

"Downtime ends here."
— UPTIME

Dayananda Sagar Academy of Technology and Management (DSATM), Bengaluru Department of Computer Science Engineering — VTU Academic Year 2026 – 2027